

Question	Marking guidance	Mark	Comments
02.1	(If) too much water the concentration of pigment (in solution) will be lower / solution will appear lighter / more light passes through (than expected); OR (If) too little water the concentration of pigment (in solution) will be greater / solution will appear darker / less light passes through (than expected); - So results (from different temperatures) are comparable;	2	1. Ignore reference to too much water so red pigment / solution too weak to measure
02.2	(Take) readings <u>s</u> (during the experiment) using a (digital) thermometer / temperature sensor;	1	
02.3	Point-to-point line drawn between co-ordinates (with a ruler); OR Smooth s-shaped line of best fit;	1	Reject <u>any</u> extrapolations below 20 °C or above 80 °C Any line should look smooth (not 'sketchy')
02.4	- Damage to (cell surface) membrane; (membrane) proteins denature; - Increased fluidity / damage to the phospholipid bilayer;	2 max	